

Step 1: In your excel spreadsheet, create a table of your distance measurements and average voltage readings.

Distance(cm)	V_avg (V)

Step 2: Select your table and go to **Insert** in the top tabs, and in the plots section select **Scatter**.

Step 3: Now that you have a scatter plot, you need to format it. You can customize your axis intervals by selecting an axis going to **Axis Options** and adjusting the **Major** units for that axis.

▼ Axis Options

Bounds

Minimum0.0↶↷

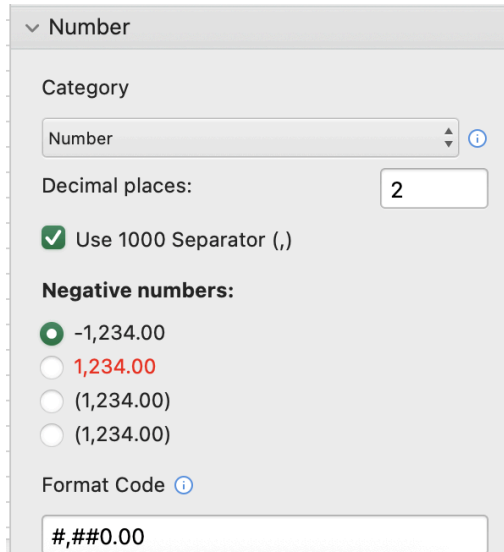
Maximum3.0↶↷

Units

Major1.0↶↷

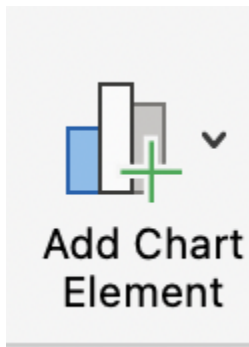
Minor0.1↶↷

Step 4: You can adjust the significant figures on your axes by selecting the axis, going down to **Number** and changing the category to **Number**. Select an appropriate amount of significant figures for your graph.



The screenshot shows a 'Number' format settings panel. At the top, there is a dropdown menu labeled 'Category' with 'Number' selected. Below this is a 'Decimal places' input field set to '2'. A checkbox labeled 'Use 1000 Separator (,)' is checked. Under the heading 'Negative numbers:', there are four radio button options: '-1,234.00' (selected), '1,234.00' (in red), '(1,234.00)', and '(1,234.00)'. At the bottom, there is a 'Format Code' field with the code '#,##0.00'.

Step 5: You can add error bars and trendlines using the **Add Chart Element** menu in the top left corner:



When you add error bars, only keep the vertical bars. When you add a trendline, choose the **best fitting trendline** for your data. At the bottom of the trendline options, be sure to check the **Display Equation** on Chart and **Display R-squared value** on chart to assess the fit of your trendline.



▼ Trendline Options

☐ Exponential

☒ Linear

☐ Logarithmic

☐ Polynomial

☐ Power

☐ Moving Average

Order

▲

▼

Period

▲

▼

Trendline Name

☒ Automatic

Linear (V_{avg} (V))

☐ Custom

Forecast

Forward

0.0

periods

Backward

0.0

periods

☐ Set Intercept

0.0

☐ Display Equation on chart

☐ Display R-squared value on chart